

One-Page Summary Report — Online Retail Dataset

Cleaning Challenges Encountered

The dataset (541,906 rows) presented several real-world data quality issues. CustomerID was missing in ~135,000 rows (25%), resolved by filling with a -1 placeholder rather than deletion, since these represented legitimate guest/unattributed transactions. Description had both missing values (1,454) and inconsistent junk entries (e.g., "damaged", "check", "wet/rusty") mixed into otherwise valid product names; these were resolved via a StockCode-to-Description mapping after excluding identified junk values, with 113 genuinely unrecoverable entries labeled "Unknown". Text formatting inconsistencies (e.g., "TOWEL, PINK" vs "TOWEL PINK") were resolved through normalization (uppercase, punctuation/whitespace cleanup). A significant discovery was that StockCode contained non-product administrative codes (POST, DOT, BANK CHARGES, CRUK, AMAZONFEE, M, S, B, D) representing postage, fees, commissions, and accounting adjustments rather than real products; these were flagged via a new IsProduct column rather than deleted, preserving them for financial reconciliation while excluding them from product-level analysis. 5,268 fully duplicated rows were identified and removed. Negative values in Quantity and UnitPrice were investigated and confirmed to represent legitimate returns, cancellations, and accounting adjustments. Three rows representing a large bad-debt write-off/reversal were removed as non-sales accounting entries. Column names were standardized to PascalCase and all text fields to uppercase for consistency.

Key EDA Findings

- **Revenue is heavily concentrated geographically:** The UK accounts for ~£8.26M of ~£9.7M total revenue, dwarfing the Netherlands, Germany, and Australia as distant secondary markets.
- **Strong seasonality:** Revenue rises sharply from September through November ahead of the Christmas period, then drops in the (incomplete) final data month.
- **Customer revenue follows a Pareto-like pattern:** A small number of high-value customers account for a disproportionate share of total revenue.
- **Growing reliance on repeat customers:** Returning customers outnumber new customers in most months from January 2011 onward, indicating strong retention but comparatively flat new customer acquisition.
- **RFM segmentation:** Identified 5 distinct customer groups (Champions, Loyal, New, At Risk, Lost), each requiring a different retention strategy.
- **Product performance varies by lens:** Top sellers by revenue are not always top sellers by quantity, and popularity (distinct order count) can differ from bulk volume.
- **Return rates vary meaningfully by product:** Most returns net out close to neutral, but specific products show disproportionately high return rates relative to units sold.
- **Purchasing shows clear time-based patterns:** Revenue concentrates on weekdays, particularly between noon and 3pm, during business hours, consistent with a wholesale/B2B-leaning customer base.
- **Non-product revenue (postage, fees, commissions)** is a small but consistent proportion of total recorded revenue across the year.

Top Insights

- **Geographic concentration risk:** with ~85% of revenue from a single country, the business has limited market diversification — international markets, while present, remain a small fraction of total sales.
- **Customer loyalty is a core strength:** Repeat customers now drive the majority of monthly revenue, but the relatively flat "New Customer" trend suggests acquisition efforts have not scaled alongside retention.
- **A small customer base drives outsized value:** The top ~20% of customers by revenue account for a large share of total revenue, making retention of "Champions" and "At Risk" segments a high-priority business action.
- **Data quality issues were operational, not random:** Nearly all "anomalies" (junk descriptions, non-product codes, extreme quantities) traced back to identifiable business processes (returns, warehouse notes, postage/fee logging) rather than corrupted data — supporting confidence in the cleaned dataset's accuracy.
- **Seasonal demand is highly predictable:** The consistent Sep–Nov revenue ramp suggests inventory and stocking strategies should be planned around this cycle rather than treated reactively.